ECO 3121 杜奕璇 12209095 Problem Set 1

Q1

1.
```
. gen yield = d32 / d31

.
. gen rent_in = c10

.
. gen rent_out = c13

. summarize yield rent_in rent_out
```

Variable	Obs	Mean	Std. dev.	Min	Max
yield	14,171	433.6169	395.1982	0	36461.98
rent_in	14,171	.5031543	4.466769	0	200
rent_out	14,171	.2164561	1.489389	0	60

The research has a total 14171 samples.
yield's data interval is [0, 36461.98] and has a mean of 433.6169 and 395.1982 std.
rent_in's data interval is [0, 200], and has a mean of 0.5032. and 4.4668 std.
rent_out's data interval is [0, 60]. and has a mean of 0.2165. and 1.4894 std.

2. β_1 will be negative: rent in land ↑ total land↑ but the totall output stays same. so yield ↓ $\beta_1<0$
β_2 will be positive: rent out land ↑ total land ↓. the total output stays same. so yield ↑ $\beta_2>0$

3.
```
. regress yield rent_in
```

Source	SS	df	MS		
Model	172005.965	1	172005.965	Number of obs	= 14,171
Residual	2.2129e+09	14,169	156180.516	F(1, 14169)	= 1.10
				Prob > F	= 0.2940
Total	2.2131e+09	14,170	156181.633	R-squared	= 0.0001
				Adj R-squared	= 0.0000
				Root MSE	= 395.2

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rent_in	-.7799978	.7432502	-1.05	0.294	-2.236866	.6768703
_cons	434.0094	3.340807	129.91	0.000	427.461	440.5578

rent_in
estimated coefficient: -0.7799978
std. error: 0.7432502
R^2: 0.0001

```
. regress yield rent_out
```

Source	SS	df	MS		
Model	13342.3767	1	13342.3767	Number of obs	= 14,171
Residual	2.2131e+09	14,169	156191.714	F(1, 14169)	= 0.09
				Prob > F	= 0.7701
Total	2.2131e+09	14,170	156181.633	R-squared	= 0.0000
				Adj R-squared	= -0.0001
				Root MSE	= 395.21

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rent_out	.6515137	2.229133	0.29	0.770	-3.717879	5.020907
_cons	433.4759	3.354809	129.21	0.000	426.9001	440.0518

rent_out
estimated coefficient: 0.6515137
std. error: 2.229133
R^2: 0.0000

The estimated value agree with my prediction

4. $\hat{\beta}_1 \approx -0.7800$

$\hat{\beta}_2 \approx 0.6515$

5.

```
. predict fitted1, xb

.
. gen residual1 = yield - fitted1
```

residual 1: mean = 2.50e-06 ≈ 0

residual 2: mean = 2.50e-06 ≈ 0

```
. predict fitted2, xb

.
. gen residual2 = yield - fitted2

. summarize residual1 residual2
```

Variable	Obs	Mean	Std. dev.	Min	Max
residual1	14,171	2.50e-06	395.197	-436.8911	36028.5
residual2	14,171	2.50e-06	395.197	-436.8911	36028.5

6. rent_in: $R^2 = 0.0001 \Rightarrow 0.01\%$ of the variation is explained

rent_out: $R^2 = 0.0000 \Rightarrow 0.00\%$ of the variation is explained

7.

```
. regress yield rental_in_share
```

Source	SS	df	MS		
Model	111081.624	1	111081.624	Number of obs	= 14,171
Residual	2.2130e+09	14,169	156184.816	F(1, 14169)	= 0.71
				Prob > F	= 0.3991
Total	2.2131e+09	14,170	156181.633	R-squared	= 0.0001
				Adj R-squared	= -0.0000
				Root MSE	= 395.2

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rental_in_s~e	.0540097	.0640427	0.84	0.399	-.0715225	.1795418
_cons	433.37	3.332743	130.03	0.000	426.8374	439.9026

$\hat{\beta}_1 = 0.0540097$

the change in yield for a one-unit increase in the proportion of rent-in land relative to the total land area, holding other variables constant

```
. regress yield rental_out_share
```

Source	SS	df	MS		
Model	3747795.36	1	3747795.36	Number of obs	= 14,171
Residual	2.2093e+09	14,169	155928.149	F(1, 14169)	= 24.04
				Prob > F	= 0.0000
Total	2.2131e+09	14,170	156181.633	R-squared	= 0.0017
				Adj R-squared	= 0.0016
				Root MSE	= 394.88

yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rental_out_~e	.2350118	.0479362	4.90	0.000	.1410505	.3289732
_cons	431.6123	3.342233	129.14	0.000	425.0611	438.1635

$\hat{\beta}_2 = 0.2350118$

The change in yield for a one-unit increase in the proportion of rent-out land relative to the total land area, holding other variables constant.

8.

```
. gen log_yield = log(yield)
(36 missing values generated)

. regress log_yield rental_in_share
```

Source	SS	df	MS	Number of obs	=	14,135
				F(1, 14133)	=	1.66
Model	.465258684	1	.465258684	Prob > F	=	0.1982
Residual	3970.8567	14,133	.280963469	R-squared	=	0.0001
				Adj R-squared	=	0.0000
Total	3971.32196	14,134	.280976508	Root MSE	=	.53006

log_yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rental_in_s~e	.0001105	.0000859	1.29	0.198	-.0000578	.0002789
_cons	5.95214	.0044757	1329.87	0.000	5.943367	5.960913

$\hat{\beta_1} = 0.0001105$

It represents the percentage increase in yield for a one-unit increase in the proportion of rent-in land to total land, holding other variables constant.

```
. regress log_yield rental_out_share
```

Source	SS	df	MS	Number of obs	=	14,135
				F(1, 14133)	=	24.13
Model	6.76799652	1	6.76799652	Prob > F	=	0.0000
Residual	3964.55396	14,133	.28051751	R-squared	=	0.0017
				Adj R-squared	=	0.0016
Total	3971.32196	14,134	.280976508	Root MSE	=	.52964

log_yield	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
rental_out_~e	.0003159	.0000643	4.91	0.000	.0001898	.0004419
_cons	5.94995	.0044885	1325.59	0.000	5.941152	5.958749

$\hat{\beta_2} = 0.0003159$

It represents the percentage increase in yield for a one-unit increase in the proportion of rent-out land to total land, holding other variables constant.

9. I prefer model(8) in 9.8. because it has a higher R^2 which means a better fit to the data.

10. SLR.4 : $E(u|x) = 0$ zero condition mean
   ① rental activities may be correlated with the error term
   ② unobserved factors like weather conditions and technological advancements.

11. ① Increase yields (Positive β_1, β_2)
   a. Access to more land : More land allows farmers to expand their cultivation area, grow more crops.
   b. Crop Rotation and Diversification : more land enable farmers to practice crop rotation, maintain soil fertility, reduce risk of pest infestations.
   ② Lower yields (Negative β_1, β_2)
   a. Resource competition : when there is a limited supply of land.
   b. Land Degradation : too many rental activities reduce soil's fertility.

Q2

1. ① Treatment indicator (T) : A binary variable that indicates whether each village was in the treatment group or the control group.
   ② Yield data (Y) : Data on village-level yields for each of the 200 villages
   ③ Covariates (X) : information on potential covariates that could affect yields or treatment assignment.

2. $Y_i = \beta_0 + \beta_1 T_i + \beta_2 X_i + \varepsilon_i$
- Y_i : village-level yield for village i.
- T_i : Treatment indicator
- X_i : Covariates that may affect yields
- ε_i : Error term

3. Not randomly : some selection ① bias may have occurred, lead to systematic differences between treatment and control groups, making it ② difficult to attribute observed differences in outcomes solely to the intervention.